

Software Documentation in a Globally Distributed Environment

Yesterday, Today, and Tomorrow

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Abstract: The Software Documentation lifecycle in a project dovetails in with the software development lifecycle. The challenges faced by the documentation team in a globally distributed setting have not received much attention. This paper highlights the challenges faced by the software documentation team in a globally distributed product development team. Further the paper elaborates on the solutions implemented, successes and failures of the team. This also includes the learnings of the team as it aligned with the Lean model of Software Development.

Keywords—globalization; software documentation; Lean model; cloud; collaboration

I. INTRODUCTION

In today's era, a term with which every organization is closely related is "Globalization". Every organization is working in a global environment producing for the globe.

With "globalization" being a keyword for today, the challenges faced during the creation of software documentation have also grown significantly. Challenges are omnipresent throughout the development of software, which in turn affects the DDLC (Documentation Development Life Cycle) too. DDLC is a well-defined methodology of various stages involved in the creation of structured software documentation. DDLC goes hand in hand with Software Development Life Cycle [1].



Fig. 1 Software Documentation Development Life Cycle

As the project teams are now spread globally, the key challenge faced by the documentation team throughout the DDLC is to communicate and collaborate with the global team to get the right information within the fixed deadline. The documentation team must also adhere to the software model for development that is followed for the project. To create quality software documentation, the establishment and support for the right infrastructure, content management system, and translation process is a mandate. Each member of the documentation team must be capable of multitasking too [2].

With the advent of the Lean model of Software development [3] within the project, the face of software documentation has also undergone a change, definitely for the better allowing us to overcome many challenges with success.

II. CASE STUDY: JOURNEY OF THE DOCUMENTATION TEAM AT EA, SIEMENS – THE HISTORY

The documentation team at EA, Siemens came into being in the year 2007 with a small team of two members. This team based at Bangalore, India, was led by the team lead who assigned tasks to the team members and discussed the tasks, status, and issues with a dedicated global documentation manager located at Germany. The documentation team mainly updated and migrated the documents and proofread the translated documents. Most of the development was carried out at other locations across the globe (for example, Nuremberg, Berlin, Vienna etc) and the development team at Bangalore had minimal communication with the globally spread development teams for the project. The documentation team based out of the locations used to review the updates and place the documents in the Content Management System (CMS). The documentation team has expanded to 11 members and has also grown in the number of projects with which it is associated.

III. CHALLENGES FACED BY THE DOCUMENTATION TEAM

To meet the twin objectives regarding on-time delivery of the documents along with quality, the documentation team encountered several challenges. Expectedly, these challenges

covered different dimensions pertaining to the processes followed, technology used etc.

A. Global coordination, communication and collaboration

This challenge is not restricted to the documentation teams but spread across other teams as well. Product teams are spread across multiple locations. Time difference among the locations is definitely a hindrance when it comes to collaboration and finding a solution to a problem faster. From a documentation perspective, to understand and author the requirements, gathering information from product teams spread across the globe is a major challenge. For example, when the review comments from the global teams are not received on time, it leads to extended work pressure later or delay in delivery. Further, due to the cultural differences, the holidays and vacations across the globe vary too. Planning and prioritizing the tasks is definitely a challenge keeping into consideration the time difference and availability of the members in order to adhere to the deadlines. Also, communicating with people from different regions with different language background is a challenge when it comes to mutual understanding of accents and language.

B. Single point of contact

The presence of a single point of contact – “Team lead” led to too much stress and workload on a single person. It was difficult for the lead to collect all the information and happenings on multiple products at a global level and pass the latest information on tasks, domain, and status for the products to all the colleagues at Bangalore in the stipulated time. In case the team lead was on emergency leave, the progress of the work was hindered. The team members had to depend on the developer/ tester and the team lead for even minor information.

C. Translation

The software products developed are released for a global audience. Hence, documentation only in English language and releasing the software documentation in a single source language does not serve the purpose. A translation process must be effectively established to translate the documents in different languages as required by the customer. Cost is one of the challenges faced in the process. Also, there are last minute updates to the documentation owing to the changes in the software. Hence, establishing a translation process that helps in capturing the last minute updates in the translated documents is a big challenge. The documentation team was dependent on both external and internal partners for the purpose owing to the challenge imposed by cost and time.

D. Adherence to the software model of development

Software documentation must always be aligned with the Software model of development. Every software model is different from each other in terms of the processes, workflow, team structure, goals, etc [4]. The changes traverse to software documentation too and the major challenge is to adapt ourselves to the changing environment accordingly. In both Waterfall and Agile model of development, there are neither

dedicated team members from the development team who provide the required support for the documentation team nor are they accountable for the time spent in support. Support from the global development team is mandatory to increase the domain and product knowledge to deliver a quality document. In the Waterfall model, the documentation team had a lot of work pressure towards the end of the release. With the advent of Lean model of software development that is a translation of lean manufacturing and lean IT principles and practices to the software development domain, the face of the challenge has also undergone a change [3]. In the Lean model of software development, the target is to produce a usable software at the end of a TAKT (TAKT is derived from the German word Taktzeit. In Lean context, it means cycle time.). Similarly, the documentation team too faced the challenge of creating a usable documentation for the software at the end of the TAKT.

E. Issues with the CMS

Together with the development team, the software documentation team is also spread across the globe. With increasing number of documents, arises the challenge of maintaining an innumerable set of documents as well as tracking the changes in the documents from different locations across the globe. Establishment of an effective content management system (CMS) [5] can definitely make the life easier. With the advent of innumerable free and licensed CMS in the market, the challenge is to select a CMS that suffices the requirements of the documentation team for content management. For licensed CMS, cost is definitely a challenge to be overcome.

In the documentation team, the support for content management system previously used was from another location. Hence, delay in support due to the difference in time zone was at times a reason for the delay in execution of the tasks. It leads to extended work pressure later or delay in the delivery. Epic editor was used as the authoring tool. To add a screenshot or to provide links to the different topics consumed almost 10 minutes. TIF graphics format was only supported. The team members had to edit the graphics, change the format, and set the resolution before adding the graphics to the document, which was leading to an extra effort to be spent on the graphics. It was not possible to copy graphics in the CMS. It would take at least 15 minutes to publish the document in any format. Software documentation was carried out across multiple locations and such instances of wastage of time often lead to ineffective global collaboration in terms of less output.

F. Issues in infrastructure for image capturing

In some cases, test systems were unavailable as the software was integrated at a later stage from various locations over the globe. This limitation did not give the documentation team the scope to view the working software beforehand to analyze, test, understand, and write or capture real time screenshots. Dependability on the developers or the testers for the same causes lead to the production of low quality graphics and wastage of time as they lack the appropriate knowledge of the screen capturing methodology.

G. Ineffective review process

The documents were sent to review to all the relevant stakeholders across the globe. This round of technical review is a mandate. An ineffective review process often leads to increased workload at the end of the release cycle. For example, the documentation team often received contradictory comments from multiple reviewers spread across different locations on the same topic making it difficult and time consuming for the author to check and address the comments on time.

H. Multitasking

A single team of technical writers were responsible for multiple projects with overlapping deadlines for delivery. There were times when a last minute request came in from the customer end to have minor changes in the template as they were viewing the document first time towards the end of the release cycle. The same change had to be encapsulated in multiple documents thereafter. Also unplanned leave of any team member required the other team members to step into his/her shoes ensuring that the delivery does not get affected.

IV. SOLUTIONS IMPLEMENTED TO MITIGATE THE CHALLENGES

A multi-pronged approach was brought in to address the challenges.

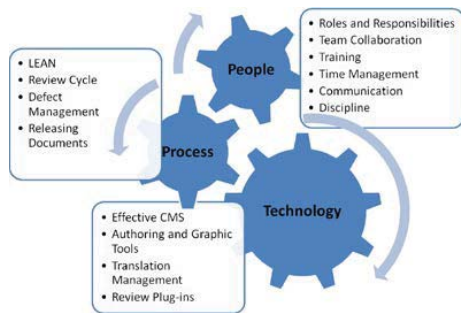


Fig. 2 Solution Implemented – People, Process, and Technology Model

With the advent of Lean model of software development, the face of software documentation for the projects aligned to it has further changed for the better.

A. Effective global communication and collaboration

The documentation team sits closely with the product development teams and is in touch with the global teams on a regular basis. The documentation team mostly interacts with teams from varying locations with time zone differences ranging from 3:30 hrs to 9:30 hrs. With the benefit of flexible timings for employees nowadays, a consensus is reached among the global team members about a range of time in which they can converse. This mode helps the documentation team in collaborating with the global product/documentation team effectively with no slippage in deadline or any additional effort. Understanding the counterparts spread across the globe

is important and touching the right chord is a key to establish team collaboration that is spread across miles. It is always good to make sure that understanding is at the same level and there exists no communication gap at all. Dedicated User Documentation representatives are present in each TAKT team. They are responsible for providing the documentation team required inputs for authoring requirements. Support from the development team is mandatory and hence it is the responsibility of the documentation team to converse with the product owners to establish the process of making them accountable for giving the required information. A template called as Functional Description for User Documentation (FDUD) comprising of questions that best answers all the questions related to a new/modified requirement must be filled by a User Documentation representative and reviewed by the User Documentation Expert and referred for authoring. Presence of both User Documentation representatives and FDUD has ensured the timely availability of information for documentation from the development teams spread globally.

B. Distribution and delegation of tasks

The responsibility and the ownership of the task were distributed among the team members. The documentation team members are empowered and the members (known as User Documentation Experts) are now a part of the planning team for their respective projects and are well informed about all the details related to the product and its delivery. User Documentation Experts are empowered to take decision on the documentation in collaboration with stakeholders spread across the globe. User Documentation Experts are the owner of the documents and are solely responsible for ensuring the quality of the documentation. If there are overlapping deadlines of multiple projects, releasing all the documents on time with high quality is the responsibility of User Documentation Expert. Tracking the status of the tasks on each project on a daily basis, delegating the tasks, and ensuring its completion is their responsibility that can be accomplished successfully with extensive planning, discipline, and time management. The team members are responsible for providing status updates to the stakeholders in uniform intervals of time. The team members directly contact counterparts based out of other locations and get their queries clarified. Dependency on developers/testers for minor issues is reduced. In case of major software documentation defects/new requirements, the team members check and analyze the issue/requirement, identify the impacted documents (locations), and provide their suggestions to developers/testers.

The documentation teams spread across globally are responsible to place the documents in the CMS and ensuring the inclusion of the documents in the software builds for the internal and external usage.

C. Transition to effective CMS and translation process

Cost is definitely a factor to establish an excellent working infrastructure. Investment in a highly usable content management system and establishment of an affordable translation management system both in terms of effort and time is definitely a value for time and money in the long run.

Also a combination of “Cloud Computing and Cloud-based authoring” [6] is a great option to establish a working infrastructure over the globe. In the cloud-based systems, all the content is centrally stored with regularly-scheduled backups and system maintenance to protect the data against accidental loss.

After several discussions, testing, and pilot-runs, the documentation team transitioned to COSIMA as the CMS. This tool supports drag and drop option allowing us to copy the graphics, text, and tables. The graphics and the links can be added to the document within 5 minutes. The tool supports single source authoring. The publication process is easy with COSIMA tool as no interface is required and it takes less time to publish in multiple output formats [7]. Dedicated support for COSIMA is also present during the Bangalore working hours leading to no wastage of time.

Translation with COSIMA is also very easy. There are dedicated translators who translate the documents present in COSIMA and translation managers who overlook the process. In case the document is not present in COSIMA, external vendors have been identified for the process.

The transition to COSIMA helped the documentation to achieve their goals on time, with more efficiency and a higher degree of overall comfort for the team.

D. Usable documentation at each TAKT

The complete software documentation planning, creation, review, translation, delivery, risk management, defect fixes, and inclusion in the build are now the ownership of the documentation team. The User Documentation Experts plan accordingly from the first TAKT itself to release a usable documentation at the end of every TAKT. Close collaboration with the global team and monitoring the progress of the development is required for the same. Extensive planning is mandatory and is the key to achieve the deadline with no compromise on quality. Seeing the unforeseen and being proactive is also very important to achieve the same. Changing the way of work to eliminate any wastage of effort is also encouraged. Once a waste is detected in any software documentation process, User Documentation Experts strive to eliminate it. For example, a User Documentation Expert has devised a process of how to develop UI messages in collaboration with the global Usability and Software documentation team thus eliminating a considerable amount of wastage of effort. In turn, the User Documentation Expert also devised a process of identifying and entering the new/updated content for translation reducing the effort of translators. Software documentation defects are also measured. It must be eradicated and is thus a criterion to be fulfilled for the product to get released globally. A defect tracking tool is used globally to analyze, check, and resolve the software documentation defects.

Software development is completed at least two TAKTS before the release so that the documentation team has ample time to incorporate last minute updates as well as review comments from all the stakeholders spread across globally on time. During the hardening TAKT, only defects are handled from the development team. Hence, there are seldom any last

minute UI changes impacting the documentation. Documents are translated after finalizing the source content. In case of minor changes to the target language documents, the User Documentation Expert updates the target language documents reducing the cost of translation. All the source and translated documents are ready two weeks prior to the release.

E. Set up of infrastructure for image capturing

To capture the screenshots, test machines are currently made available to the documentation teams at all the locations on time. The graphic file formats supported are TIF and PNG format. There is no need to set the resolution of the graphics to place it in the document. In case of non-availability of real-time screenshots in certain cases, the team still depends on testers.

F. Setting up of effective review process

Each type of review (self review, peer review, editorial review, and technical review) is planned diligently to create a high quality document with zero defects. Deadlines for any review are communicated clearly and beforehand to all the stakeholders to ensure the timely completion. A reminder is also sent to the stakeholders a day or two before the deadline to ensure its completion. Each documentation team assigned for the project plans interim reviews to avoid too many last minute review comments. For the review, the team is using a review plug-in. As a result, all the reviewers spread across globally are able to view the comments provided by each other and hence avoid duplicate comments from different authors. It takes less time for the author to incorporate the comments.

V. LESSONS LEARNT

Implementing Lean and improvising the existing processes definitely helped the documentation team in going miles ahead reinventing their roles and responsibilities and in establishing an effective collaboration with the global team. It resulted in the ownership of software documentation (targeted for the global audience) for multiple projects at Bangalore.

A. What went well

- Transparency with the global teams
- Being proactive
- Global responsibility of software documentation for multiple projects at Bangalore
- Adherence to the Lean model
- Distribution and delegation of tasks across the team members over the globe
- Introduction/Improvement of processes and technology
- Increase in domain competency
- On-time delivery
- Availability of documents globally

B. What did not go well

- Non acceptance of the huge number of changes in process and technology by certain team members
- Time taken to prove the efficacy of the Bangalore team to the global team to establish effective collaboration
- Time taken to adapt the new processes and technologies

C. Lessons learnt

Establishing processes that support efficient software documentation and embracing technology that eases the effort and management together helped in creating quality software documentation in a globally distributed environment.

The following figure depicts the growth of the documentation team at EA, Siemens:

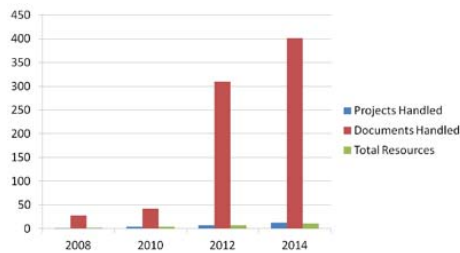


Fig. 3 Journey – From 2008 till date

VI. CONCLUSION

The paper can be summarized as the amalgamation of the well-defined solutions and approach that was undertaken to develop software documentation efficiently. The solutions have worked out effectively with success. The product documentation teams can definitely identify with the challenges and the solutions that can be incorporated into their respective documentation processes.

Change is something that is constant. Hence, the team cannot stop at the above mentioned solutions. There might arise a new set of challenges going forward, hence the learning is to constantly improvise according to the changing challenges and environment to overcome the same always in due course of time. There is no dearth of learning.

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